B.Tech II Year (3rd Semester)

AMXXXXX **SOLID MECHANICS** **CREDIT: 05 (3-1-2)**

UNIT 1. Elements of Engineering Mechanics 08 (L)

Idealization in Mechanics, Free Body Diagrams and Equilibrium of Force Systems, Application to Structures: Analysis of Plane Trusses

Geometric Properties of Sections: C.G. and Centroid, Area Moments of Inertia, Transformation of Moments of Inertia and Principal Axes

UNIT 2. Analysis of Stresses and Strains: 08(L)

Uniaxial Stress and Strain: Stress, Strain, Hooke's Law, Stress-Strain Curves, Elastic Constants, Strain Energy, Statically Indeterminate Problems, Thermal Effect, Impact Loading.

Biaxial Stress and Strain: Stress at a Point, Stress Transformation and Principal Stresses, Analysis of Strain, Strain-Displacement Relations, Constitutive Equations, Strain Transformation and Principal Strains, Strain Measurement.

UNIT 3. Stresses in Beams and Torsion of Shafts 08(L)

Beams: Loading vs. Shear Force vs. Bending Moment Relations, S.F.D and B.M.D, Pure Bending, Normal and Shear Stresses, Combined Bending and Axial Stresses, Composite Beams

Shafts: Torsion of circular Shaft, Power Transmitted by a Shaft, Compound Shaft, Stain Energy in Torsion, Combined Bending and Twisting; Strain Energy in Bending and Torsion

UNIT 4. Deflections of Beams 06(L)

Equations of Elastic Curve, Methods of Double Integration for Determining Deflections, Macaulay's Method, Moment-Area Method, Castigliano's Theorem

UNIT 5. Columns and Theories of Failure: 06(L)

Euler 's Theory for long columns, Rankine-Gordon Formula, Eccentrically Loaded Columns, Theories of Failure

List of Practicals

1. Verification of conditions of equilibrium for concurrent (polygon law) and parallel forces and moment systems
2. To determine the coefficient of friction in case of rollers and sliding boxes over inclined planes
3. To determine the mass moment of inertia of a fly-wheel
4. Experimental verification of forces in a roof truss model
5. To determine the mechanical properties of mild steel specimen on a universal testing machine (UTM).
6. To perform the torsion test on a given specimen.
7. To determine the value of modulus of elasticity by measurement of slope and deflection of a beam using beam bending apparatus.
8. To determine the impact strength of the material using Charpy test and Izod test.
9. To determine the maximum shear stress (parallel to grains) of timber.
10. To study the behavior of different types of struts and calculation of Euler's buckling load in each case

Text/References Books:

1. Beer F.P. and Johnston E.R., Mechanics of engineers-volume 1-Statics, Volume II – Dynamics,McGraw Hill,New York.
2. Merriam J.L and Kraige L.G. Engineering Mechanics,Volume I-Statics,Volume-II-Dynamics, John Wiley & Sons New York
3. Shames L.H. Engineering Mechanics,PrenticeHall,New Delhi
4. R.C. Hibbler, Engineering Mechanics, Vol I and II,Pearson Press,2002
5. Elements of strength of materials, S.P. Timoshenko and D.H. Young, East- West Press Pvt.Ltd.Publication
6. Mechanics of Materials,Pytel and Kiusalaas,Cengage Learning Publication.
7. Mechanics of Materials,Gere and Timoshenko,CBS Publications.
8. Mechanics of Materials, E.P. Popov, Prentics Hall Publications.
9. Hibbeler, R.C., Mechanics of materials. 10th Edition Pearson Educación.
10. Strength of Materials,G.H. Ryder, Macmillan India Limited.
11. Strength of Materials- Pytel and Singer, HarperCollins College Division Publications.
12. Strength Of Materials, Crandals,Dahal and Lardener, Tata McGraw Hill Publications.
13. Mechanics Of Materials – Riley, Struges and Morris,John Wiley & Sons

CE XXXX: CONCRETE TECHNOLOGY AND CONSTRUCTION MANAGEMENT

CREDIT: 04 (3-0-2)

Unit 1: Properties of Concrete Materials **(6L)**

Composition; Physical & chemical properties of cement; Testing of cements as per BIS specifications; Types of cements, admixtures, and aggregates; Effects of plasticizers and superplasticizers

Unit 2: Fresh Concrete & Hardened Concrete **(12L)**

Concrete mix design, Workability and its measurements; Compaction & curing of concrete; Strength of concrete; Factors affecting strength of concrete; Micro structure of hardened concrete; NDT & Conventional testing of concrete; Light weight concrete; High strength concrete; Roller compacted concrete; Self compacted concrete; High performance concrete.

Unit 3: Durability of Concrete **(6L)**

Mechanism of deterioration of concrete; Different modes of transport mechanism in concrete mass; Carbonation; Chloride attack; Corrosion of reinforcement in concrete; Carbon economics; Life cycle costing and energy considerations.

Unit 4: Construction Engineering Systems **(6L)**

Introduction to construction projects–magnitude and critical considerations; Bar charts, milestone charts, and network diagrams; Resource quantification; CPM, PERT, PDM and GERT; Critical path evaluation; Cost planning and resource allocation through network techniques; Project monitoring and controls; Line of balance technique.

Unit 5: Engineering Economics & Construction Contracts and Delivery **(6L)**

Time value of money; Present economy studies; Equivalence concept; Economic comparison; Depreciation and break even cost analysis; Contract and contract management; Tenders and tender preparation; settlements of disputes, arbitration and commissioning of project; Decision making under risk–decision tree concept.

List of Practical

1. To determine the specific gravity of cement using Lechatelier flask and the fineness by sieve analysis.
2. To determine the normal consistency and setting times.
3. To assess the soundness of cement using Lechatelier apparatus.
4. To determine the compressive strength of cement.
5. To determine the specific gravity, bulk density and water absorption of aggregates.
6. To study the phenomenon of bulking of sand
7. To draw the grading curves for fine and coarse aggregates and hence to determine their fineness moduli.
8. To determine the crushing value, impact value and ten percent fine value for coarse aggregates.
9. To measure the workability of concrete using slump cone, and compaction factor apparatus
10. To design a concrete mix of given specifications and to evaluate associated trial mixes.

References

1. M. Neville, Properties of Concrete, Pearson education.
2. P. K. Mehta & Paulo J. M. Monterio, Concrete: Microstructure, Properties, and Materials, Tata Mc Graw Hill.
3. M. S. Shetty, Concrete Technology, theory and Practice, S. Chand
4. M. L. Gambhir, Concrete Technology, Theory and Practice, Mc Graw Hill.
5. J. Kroop and H. K. Hilsdorf, Performance Criteria for Concrete Durability, E & F N Spon, London.
6. Panneerselvam, R., Engineering Economics, Prentice- Hall of India Private Limited New Delhi.
7. Seetharaman, S., Construction Engineering and Management, Umesh Publications New Delhi.
8. Project Planning and Control with PERT and CPM Book by B.C. Punmia and K. K. Khandelwal

AMXXXXX FLUID MECHANICS & HYDRAULICS CREDIT: 05 (3-1-2)

UNIT 1. Introduction, Fluid Statics and Kinematics (04)

Fluid and continuum, Physical properties of fluids, Types of fluid flows, , manometers, pressure transducers, pressure on plane and curved surfaces, centre of pressure. Kinematics of Fluid flow: steadiness, uniformity, rotational and irrotational flows, streamline, streakline, pathline, continuity equation, stream function and velocity potential, flow net.

UNIT 2. Dynamics of Fluid Flow, Dimensional Analysis and Bundary Layer (13)

Euler's Equation of motion along a streamline and its integration, Bernoulli's equation and its applications, momentum equation and its application to pipe bends.

Dimensional Analysis. Equation of motion for laminar flow through pipes, transition from laminar to turbulent flow, types of turbulent flow, velocity distribution in turbulent flow over smooth and rough surfaces, resistance to flow, minor losses, pipe in series and parallel, power transmission through a pipe, water hammer (rigid theory). Introduction of boundary layer, displacement and momentum thickness, boundary layer over a flat plate.

UNIT 3. Basic Open Channel Hydraulics (05)

Classification, types and regimes, Channel geometry. Velocity and momentum distribution, Specific energy and Critical flow, Specific force. Qualification of Uniform flow, Velocity measurement, Chezy's and Manning's Equation. Determination of Roughness Coefficient, Normal Depth and Velocity. Most Economical Section, Non-erodible Channels. Flow in channel with equivalent roughness, Horton's method of Equivalent, roughness estimation, Semi-filled flow in close conduits.

UNIT 4. Advanced Open Channel Hydraulics (08)

Dynamic equations of gradually varied flow, Draw-down and Back water curves, Channel slopes, Classification of surface profile, Graphical integration, Direct and standard step method, Numerical methods, Flow through channel transitions, Dynamic equation of spatially varied flow, Spatially varied flow profile, Computation of spatially varied flow. Hydraulic jump, Types and analysis of hydraulic jump, Basic characteristic of jump, Length, Profile jump, Jump on sloping floor, Location of jump, Jump as energy dissipation, Control of jump, Surges in open channel. Sudden transition, Sub-critical flow through sudden transition, Flow through culverts, Flow through bridge piers, High flow through bridge pier and obstruction, Channel junction.

UNIT 5. Hydraulic Machines (06)

Hydraulic Turbines: Introduction to Hydroelectric power station and its components, Classification of turbines, Heads and efficiencies, Pelton wheel, Francis and Kaplan turbines, specific speed and unit quantities, Characteristic curves. Pumps: Centrifugal Pumps, specific speed, priming, Characteristic curves, Reciprocating pumps, Comparison between Centrifugal and Reciprocating pumps. Cavitation in pumps and turbines.

List of Experiments:

1.To verify the momentum equation using the experimental set-up on diffusion of submerged air jet.
2. To study the transition from laminar to turbulent flow and to determine the lower critical Reynolds number.
3. To study the boundary layer velocity profile over a flat plate and to determine the boundary layer thickness.
4. To calibrate an orifice meter, venturimeter, and bend meter and study the variation of the co-efficient of discharge with the Reynolds number.
5. To study performance of a Centrifugal pump / Reciprocating pump.
6. To study the impact of jets on a flat plate.
7. To study performance of a Turbines (Pelton / Francis / Kalpan).
8. Discharge of an open channel using rectangular / V-notches.
9. To study the characteristics of hydraulic jump in an open channel.

References:

1. *Som, S.K. and Biswas G, Introduction of Fluid Mechanics & Fluid Machines, TMH, New Delhi.*
2. *Fox, R.W., McDonald, A.T., Introduction to Fluid Mechanics, 7th edition, Wiley India.*
3. *Ojha, C.S.P., Berndtsson, R., Chandramouli, P.N., Fluid Mechanics and Machinery, Oxford University Press, New Delhi.*
4. *Ranga Raju, K.G., Open Channel Hydraulics, , Tata McGraw Hill, 2nd Ed.*
5. *Garde, R.J., Fluid Mechanics through Problems, New Age International Pvt. Ltd, New Delhi.*
6. *Majumdar, B., Fluid Mechanics with Laboratory Manual, PHI Learning, New Delhi.*

CEXXXXX **ENGINEERING GEOLOGY** **CREDIT: 03 (2-0-2)**

UNIT 1. Rocks **06(L)**

Minerals: Physical and Engineering properties of rock. Origin, structure, texture and classification of igneous, sedimentary and metamorphic rocks and their suitability and their impacts in Civil Engineering projects.

UNIT 2. Geological Structures **05(L)**

Rock deformation**:** Folds, Faults, joints unconformity and their applications in Civil Engineering.

UNIT 3. Earthquake **02(L)**

Earthquake, its causes, classification and seismic zones of India.

UNIT 4. Landslide and Rain Water Harvesting **05 (L)**

Landslides, its causes, classification and preventive measures with a case study. Concept, techniques and design of Rainwater harvesting with case study.

UNIT 5. Geological Investigations **06 (L)**

Modern approach under geological investigations for site selection of Dams and reservoirs, tunnels, bridges and Highways.

References:

1. *Krynine and Judd : Principal of Engg. Geology and Geotechnics, Mc Graw Hill Book Company Inc. New York.*
2. *J.M. Treteth : Geology of Engineers, Princeton, Von. Nostrand.*
3. *Prabin Singh : Engg. and General Geology, Katson Publishing House.*
4. *Blyth F.G.M. : A Geology for Engineers, Arnold, London.*
5. *D.S. Arora : Geology for Engineers, Mohindra Capital Publishers, Chandigarh.*
6. *Leggot, R.F. : Geology and Engineering, McGraw Hill, New York.*
7. *P.K. Mukerjee : A text Book of Geology, Calcutta Word Publishers.*
8. S.K.Duggal, H.K.Pandey, NR Rawal, Mc-Gra Hill Publisher

List of Practicals

1. Study of physical and engineering properties of Quartz, Felspar, Calcite and Mica Minerals. **(4hrs)**

2. Study of physical and engineering properties of Granite, Basalt, Sandstone, Marble, Quartzite, Limestone, Granite Gneiss, Mica Schist and their applications in Civil Engineering structures/Projects. (**8hrs**)

3. Petrography of selected minerals and rocks through thin sections under Trinocular Polarizing Optical microscope with reference to alkali silica reaction and textures. (**4hrs)**

4.Study of different geological models and their relevance to the different engineering structures (**6 hrs)**

5. Preparation of cross (X) sections of different geological maps including calculation of dip and its direction. **(2hrs)**

References:

1. *Krynine and Judd : Principal of Engg. Geology and Geotechnics, Mc Graw Hill Book Company Inc. New York.*
2. *J.M. Treteth : Geology of Engineers, Princeton, Von. Nostrand.*
3. *Prabin Singh : Engg. and General Geology, Katson Publishing House.*
4. *Blyth F.G.M. : A Geology for Engineers, Arnold, London.*
5. *D.S. Arora : Geology for Engineers, Mohindra Capital Publishers, Chandigarh.*
6. *Leggot, R.F. : Geology and Engineering, McGraw Hill, New York.*
7. *P.K. Mukerjee : A text Book of Geology, Calcutta Word Publishers.*
8. S.K.Duggal, H.K.Pandey, NR Rawal, Mc-Gra Hill Publisher

CEXXXXX **SURVEYING** **CREDIT: 05 (3-1-2)**

Pre-Requisite: Building Engineering-I

Unit- 1: Course Fundamentals, Traversing & Trilateration **(7 L)**

Introduction to the course; Traversing- Fundamentals & methods, traverse computations- latitude & Departure, Traverse Adjustment, Gales Traverse Table, Omitted Measurements; Total Station; Trilateration- Principle and methods.

Unit- 2: Trigonometrical Levelling, Contouring & Tacheometry **(8 L)**

Review of levelling concepts; Trigonometrical Levelling- observations in single vertical plane; Contours- Characteristics, methods and uses; Tacheometry: Basic terms & principles, Distance elevation formula, Stadia method.

Unit- 3: Triangulation and Plane Table Surveying **(6 L)**

Triangulation- Principle and classification, Triangulation field work, intervisibility of stations, satellite station; Plane Table Survey- Principles, accessories, methods- radiation, intersection & traversing, Resection by three point problem, Lehmann's rules.

Unit- 4: Curve Surveying: Simple, Compound, Reverse, Transition & Vertical **(9 L)**

Curve Surveying- Elements of simple circular curves, Setting out simple circular curves by linear & instrumental methods; Compound curves- Elements and setting out; Reverse curves; Transition curves- Need and characteristics, Ideal transition curve and intrinsic equation, Transition curves and equations, Combined transition curve and circular curves- shift, elements and setting out; Vertical curves- Types, Methods of setting out.

Unit- 5: Theory of Errors and Adjustment Computations **(6 L)**

Errors, Accuracy and precision, Characteristics of random errors, Weighting of observations, Propagation of errors, Principle of least squares, Most probable value, Determination of mpv, Method of normal equations, Method of differences, Observation equations accompanied by condition equation, Adjustment of simple triangulation figures.

List of Practicals

1. To perform fly levelling using automatic level.
2. To measure horizontal angle between four objects by method of reiteration using a Vernier theodolite.
3. To determine tacheometric constants of a Vernier theodolite.
4. To set out a simple circular curve by Rankine's method of tangential angles.
5. To study working of total station and practice for taking observations.
6. To prepare topographic map of small area using total station (Day-01).
7. To prepare topographic map of small area using total station (Day-02).
8. To carry out control surveying [using total station] and plot the coordinates of control points at a given scale on Plane Table and their field checking.
9. To solve three point problem in Plane Table surveying and plotting of details in an area.
10. To plot details of an area using radiation method.
11. To plot the details of an area using intersection method.
12. Colouring of map prepared by Plane Table surveying and submission.

References:

1. Surveying (Vol- I & II): B.C. Punmia & A.K. Jain, Laxmi Publications, New Delhi.
2. Surveying (Vol- I &, II): S.K. Duggal, Tata McGraw-Hill, New Delhi.
3. Surveying (Vol- I & II): K.R. Arora, Standard Book House, New Delhi.
4. Plane Surveying & Higher Surveying: A.M. Chandra, New Age Int. (P) Ltd., New Delhi.
5. Surveying: Bannister, A., Raymond S., Baker, R., Pearson Education, New Delhi.

II Year (4th Semester)

CEXXXXX WATER SUPPLY ENGINEERING CREDIT: 04 (3-1-0)

UNIT 1. Water Demand and Sources (6L)

Introduction, population forecasting, variations in water demand, estimation of quantity of water. Sources of Water: determination of capacity of impounding reservoirs: Suitability of surface and subsurface sources; Rain water harvesting intake structures.

UNIT 2. Water Quality parameters (6L)

Necessity of water quality examination, examination of various water quality parameters including chemical interactions, water quality standards for various beneficial uses.

UNIT 3. Pumping and conveyance of water (5L)

Pipes for conveyance of water, hydraulics of pressure and gravity conduits, capacity and sizes including economical size of rising main, pipe laying, types of pumps, selection of pumps, characteristics curve, design of pumps.

UNIT 4. Water Distribution System (6L)

Layouts and components of water distribution system, Method of water supply and storage, principle of analysis of distribution network-Hardy Cross method, Equivalent pipe method. Application of software for network analysis.

UNIT 5. Conventional and Miscellaneous treatment of water (13L)

Objective of water treatment, Water treatment Unit Operation and Unit Processes, Chemistry and microbiology of different water treatment systems, Water treatment flow sheets.

Sedimentation: Theory of sedimentation, Efficiency of ideal and real settling tank, settling and removal efficiency of discrete and flocculant particles. Design of primary sedimentation tanks & tube settlers etc.

Coagulation: Theory & mechanism of coagulation, Coagulant and their reaction, Determination of optimum dose of coagulant and design of rapid mixer.

Flocculation: Theory of flocculation, Hydraulics and mechanical flocculator and their design. Design of secondary sedimentation tank.

Filteration: Theory of filtration, Hydraulic of filtration and backwashing slow sand and rapid sand filters, brief introduction to dual & multi media filters, deep filteration

Disinfection: Theory and mechanism of disinfection, various disinfectants and their application, chlorination and practice of chlorination & dose estimate.

Water Softening: Membrane filtration, Miscellaneous treatment, removal of priority pollutant.

References:

1. McGhee: Water supply and sewerage, Tata McgrawHill, publication.
2. Peavy, Rowe and Techbanoglous: Environmental Engineering, Tata McgrawHill, publication.
3. MetCalf & Eddy: Wastewater Engineering: Treatmment and Reuse, Tata McgrawHill, publication.
4. Sawyer and McCarty: Chemistry for Environmental Engineering, Tata McgrawHill, publication.
5. Garg S. K.: Environmental Enggineering (I&II),Khanna publication, New Delhi.
6. B.C. Punamia & Jain A.: Environmental Enggineering (I&II), Laxmi publication, New Delhi.
7. Weber, W.J., “Physico-chemical Processess”, Wiley Interscience 1983
8. S. Vigneswaran and C. Visvanathan, “Water Treatment Processes: Simple Options”, CRC Press. 1995
9. R.L.Droste, “Theory and Practice of Water and Wastewater Treatment”, John Wiley. 1997
10. S.R. Qasim, Edward and Motley and Zhu, H., “Water Works Engineering – Planning, Design and Operation”, Prentice Hall, India. 2002
11. Nicholas G. Pizzi, “Water Treatment Operator Handbook”, American Water Works Association. 2005

CEXXXXX **STRUCTURAL ANALYSIS- I** **CREDIT: 04 (3-1-0)**

UNIT 1. Introduction & Analysis of Plane Structures **12(L)**

Introduction and Classification of Structures, Static and Kinematic Indeterminacy of Structures, Analysis of Compound and Complex Trusses, Analysis of Plane Frames. Energy methods, Maxwell's Reciprocal & Betti's Theorem, Unit Load method, Deflection of trusses and plane frames.

UNIT 2. Rolling Loads and Influence Line Diagrams **07(L)**

Introduction, Influence Line Diagrams for Beams & Trusses, Absolute Maximum Bending Moments, Muller-Breslau principle and its applications.

UNIT 3. Arches, Cables and Suspension Bridges **07(L)**

Introduction, Linear Arch, Eddy's Theorem, Three-Hinged & Two-Hinged Arches, Spandrel Braced Arch, Influence Lines for Arches, Analysis of Cables, Suspension bridges with three and two hinged stiffening girders.

UNIT 4. Unsymmetrical Bending **04(L)**

Introduction, Location of Neutral axis, Computation of Stresses and deflections.

UNIT 5. Curved Beams **06(L)**

Bending of curved beams in plane of bending, stresses in bars with small and large initial curvatures, Beams Curved in plan.

References:

1. Structural Analysis, Hibbeler, Pearson Publications.

2. Structural Analysis, Aslam Kassimali, Cengage Learning Publications.

3. Structural Analysis in Theory and Practice, Alan Williams, Elsevier Publications.

4. Elementary Structural Analysis, C. H. Norris, J. B. Wilbur and S. Utku, Tata Mcgraw Hill Publications.

5. Structural Analysis, L.S. Negi and R. S. Jangid, Tata Mcgraw Hill Publications.

CEXXXXX HIGHWAY & TRAFFIC ENGINEERING CREDIT: 04 (3-1-0)

UNIT 1. Highway Planning and Alignment 05(L)

Introduction to Highway Development; Institutions for Highway Development at National level; Highway Planning and Alignment; Engineering surveys for highway planning. Vision Documents for Rural and Urban Roads (2021, 2025, NUTP). Classification of Roads, Network patterns.

UNIT 2. Highways Geometric Design 08(L)

Highway Cross Sectional Elements. Factors affecting geometric design. Sight Distances. Design of Horizontal and Vertical Alignments. Geometric Design considerations in hilly areas.

UNIT 3. Traffic Characteristics 07(L)

Road user, vehicle and highway characteristics. Traffic Stream Characteristics- Macroscopic, Microscopic. Distance-Time diagram. Traffic Flow Diagrams, Highway Capacity, Level of Service.

UNIT 4. Traffic Studies & Design 08(L)

Traffic Studies- Volume studies, Speed studies, Origin & Destination studies, Parking & Accident studies; Design of rotary and signalized intersections. Signal Coordination.

UNIT 5. Traffic Control 08(L)

Traffic Signs & markings, Islands, Signals; Intersections: At grade and grade separated intersections, rotary intersection; Traffic Regulation measures, Road Safety.

References:

1. Highway Engineering by S.K. Khanna and C.E.J. Justo, Nem Chand Publication.
2. Principles of Transportation Engineering, by Animesh Das and P. Chakraborty, PHI Learning; 2nd edition
3. Traffic Engineering and Transport Planning by L.R. Kadiyali, Khanna Publishers.
4. Indian Roads Congress [IRC] Codes –IRC:3, 38, SP-48, 69, 73, 86, SP-23, SP-44, SP-88

CEXXXXX GEOTECHNICAL ENGINEERING – I CREDIT: 04 (3-0-2)

Prerequisites: Strength of Materials

UNIT 1. Soil formation, Properties and Classification of soils 10(L)
Soil mechanics and its importance, Particle size analysis, Phase relationship, Index properties, Soil mineralogy and structure, Identification and Classification of soils, Soil Classification systems.

UNIT 2. Permeability and Capillarity 07L)
Types of soil water, effective stress principle, Permeability and seepage of soils, Coefficient of permeability and its determination in laboratory and field, Quick sand and Liquefaction phenomenon, Seepage analysis, Flow nets and its construction, Seepage through earthen embankments.

UNIT 3. Compaction and Consolidation of Soils 10(L)
Theory of compaction, Standard and Modified Proctor test, Effect of compaction on properties of soils, Field compaction of soils, Compaction control in fields.
Compressibility of soils, Types of consolidation, Terzaghi's theory of consolidation, Consolidation test, Determination of coefficient of consolidation, Pre-consolidation pressure and its determination, Time rate of consolidation, Computation of settlement, Sand drains.

UNIT 4. Shear Strength of Soils 05(L)
Mechanism of shear resistance, Mohr-Coulomb theory, Shear strength and effective stress principle, Shear tests under different drainage conditions, Pore pressure parameters, Shear characteristics of cohesion less and cohesive soils, Modified failure envelop, Stress path.

UNIT 5. Stability of Slopes 04(L)
Types of slope failure, Stability of infinite and finite slopes, Taylor's stability number, Stability analysis – Culmann's method, Swedish Circle method, Friction circle method, Bishop's method, Stability charts.

List of Experiments:
1. Visual identification and specific gravity
2. Grain Size Analysis: Sieve Analysis & Hydrometer Analysis
3. Atterberg's Limits
4. Relative Density Test
5. Proctor Compaction Test
6. In site Density-Core Cutter & Sand Replacement Method.
7. Permeability Test: Constant Head & Variable Head
8. Direct Shear Test
9. Shear Strength Test: Unconfined Compression Test & Triaxial test
10. Consolidation test

***References**:*

1. *Lambe & Whitman: Soil Mechanics, Wiley-India.*
2. *Gopal Ranjan and A.S.R. Rao: Basic and Applied Soil Mechanics, New Age International*
3. *B. M. Das: Principles of Geotechnical Engineering, CL-Engineering.*
4. *D.F. Mc Garthy: Essentials of soil mechanics and foundation. New Age International*
5. *Alam Singh: Modern Geotechnical Engineering., CBS publisher*
6. *K.R. Arora: Soil Mechanics and Foundation Engineering, Standard Publishers Distributors*
7. *Purushotama Raj: Geotechnical Engineering, New Age International Limited*
8. *SP 36-1: Compendium of Indian Standards on Soil Engineering: Part-1 Laboratory Testing of Soils for civil Engineering Purposes.*

CEXXXXX **GEOINFORMATICS** **CREDIT: 04 (3-1-0)**

Pre-requisite: Nil

Unit - 1: Remote Sensing (8 L)

Introduction to Geoinformatics, Physics of remote sensing, remote sensing system, Spectral reflectance curves, resolution and multi-concept, Sensors and orbital characteristics, Visual interpretation- FCC, Remote Sensing satellites and data products.

Unit - 2: Satellite Image Processing (7 L)

Satellite Image- Characteristics and formats, Image statistics, Land use and land cover classification system, Image rectification, Unsupervised and Supervised classification.

Unit - 3: Geographic Information System (GIS) (8 L)

Geographic data- Basic concepts, GIS and its components, Data acquisition, Raster and Vector models, topology, creation of integrated geographic database, Data organization, Query formation, Buffering, Spatial Overlay, Data output.

Unit - 4: Global Positioning System (GPS) (6 L)

Introduction, Satellite navigation System, Datums and Map projections, GPS- Basics and segments, GPS signal, Static, Kinematic and Differential GPS, NavIC fundamentals.

Unit - 5: Applications and Open Source Software (OSS) (7 L)

Applications of remote sensing, image processing, GIS and GPS in Civil Engineering and related fields; OSS, Open data and Open standards, Hands on practice on OSS in the field of GIS; Hands on practice on OSS in the field of image processing.

Reference Books:

1. *Remote Sensing & Image Interpretation: T.M. Lillesand & R.W. Kiefer; Wiley Students Edition, India.*
2. *Principles of Geographical Information Systems: Peter A. Burrough & Rachael A. McDonnell; Oxford University Press.*
3. *Principles and Theory of Geoinformatics: P.K. Garg; Khanna Book Publishing, Delhi.*
4. *GIS and Remote Sensing: A.M. Chandra & S.K. Ghosh; Narosa Publications, Delhi.*
5. *Computer Processing of Remotely Sensed Images: P.M. Mather; John Wiley & Sons.*
6. *Remote Sensing and GIS: M. Anji Reddy; BS Publications, Hyderabad.*
7. *Essentials of GPS: N.K. Agarwal; Spatial Networks Publishers, Hyderabad.*

III YEAR (5th SEMESTER)

CEXXXXX PAVEMENT ENGINEERING & MANAGEMENT CREDIT: 04 (3-0-2)

Unit 1: Pavement Types, materials and their characterization (07 L)

Pavement Types and their classification – Flexible, and Rigid pavements, Highway and low volume roads and their pavement types; Pavement components, Highway Construction materials and their Characterisation – soil, aggregate, bitumen, cement and unconventional materials, Material Characterizations/ specifications as per relevant codes, Cementitious materials: Cement treated base; cement treated sub-base; stabilised soil, characterization, Bituminous mixes: Types of mixes; Marshall mix design; Mix specifications, Emerging Sustainable Materials for pavement construction.

Unit 2: Flexible Pavement Design (07 L)

Stresses in flexible pavements: layered system concepts, stress solution for one, two, and three-layered systems, fundamental design concepts; Factors effecting pavement design and variables considered in pavement design: axle types, standard and legal axle loads, ESWL, EWLF, vehicle damage factor, ADT, AADT, growth factor, lane distribution factor, directional distribution factor, tire pressure, contact pressure, design life; Mechanistic Methods of flexible pavement design, Drainage considerations, design of flexible pavements of low volume roads, design of flexible pavement using IRC method and AASHTO method

Unit 3: Rigid Pavement Design (06 L)

Stresses in rigid pavements: Westergaard's theory and assumptions, stresses due to curling, stresses and deflections due to loading, frictional stresses, design of joints; design of rigid pavement using IRC method, Design of Rigid Pavement Thickness as per IRC Provisions, Design of Dowel bars, Design of Tie bars, Design factors for Runway pavements, Design methods for Airfield pavements.

Unit 4: Construction Methods of Pavements (10 L)

Types of bituminous constructions, Construction equipment, Construction procedure for embankments, subgrades, subbase, drainage layer, filter /separation layers and base courses-WBM, WMM, stabilized layers, Interface treatments, Constructions practices for Binder & Wearing Courses for roads as per IRC and MORT&H specifications, IRC Specifications for Dry lean concrete, Pavement quality concrete, construction procedure and requirements for rigid pavements as per IRC and MORTH specifications, Construction methods for joints in rigid pavements as per IRC Provisions, Quality Control in Construction

Unit 5: Highway Maintenance (06 L)

Introduction, Pre-requirements, organizational setup, specification and code of practices, Distresses/Defects in flexible and rigid pavements, Structural and functional evaluation, Pavement Evaluation Methods, Types of Maintenance Activities, Overlay Design as per IRC Provisions.

References:

Books:

1. Chakroborty, P., & Das, A. Principles of transportation engineering. PHI Learning Pvt. Ltd.
2. Huang, Y. H. Pavement analysis and design, Upper Saddle River, NJ: Pearson Prentice Hall.
3. Islam, M. R., & Tarefder, R. A. Pavement Design. Pavement Design: Materials, Analysis, and Highways. McGraw-Hill Education.
4. Kadiyali L. R. and Lal, N. B., Principles & Practice of Highway Engineering, Khanna Publishers,
5. Kandhal, P. S. Bituminous road construction in India. PHI Learning Pvt. Ltd..
6. Khanna, S. K., Justo, C. E. G., & Veeraraghavan, A. , Highway Engineering, Nem Chand & Bros Publishers.
7. Sharma, S. K. Principles, practice and design of Highway Engineering. S. Chand Publishing.

Codes:

1. IRC 82-2015, Code of practice for maintenance of bituminous surfaces of highways
2. IRC: SP:83-2015, Maintenance & Rehabilitation of Cement Concrete Pavements
3. IRC: 37-2018, Guidelines for the design of flexible pavement
4. IRC: 58-2015, Guidelines for the design of plain jointed rigid pavement for highways
5. IRC:15 -2017, Standard specification and code of practice for construction of concrete roads
6. IRC:120, Recycling of Bituminous Pavements
7. IRC SP-100, Use of Cold Mix Technology in Construction and Maintenance of Roads Using Bitumen Emulsion
8. IRC:82, Code of Practice for Maintenance of Bituminous Surfaces of Highways
9. MoRTH (Ministry of Road Transport and Highways). (2013). Specifications for road and bridge works. In Indian roads congress. New Delhi, India: MoRTH.
10. IRC:81, Guidelines. for Strengthening of Flexible Road Pavements using Benkelman beam deflection
11. Other relevant IRC & IS Codes

List of Practical:

1. Tests on Aggregate:
   (a) Impact Test.
   (b) Abrasion Test.
   (c) Shape Test.
2. Tests on Bituminous materials:
   (a) Penetration Test.
   (b) Viscosity Test.
   (c) Ductility Test.
   (d) Stripping Value Test.
   (e) Softening point Test.
   (f) Flash and Fire point Test.
3. Marshall Mix Design
4. Traffic studies:
   (a) Traffic Volume study.
   (b) Traffic speed studies.

CEXXXXX GEOTECHNICAL ENGINEERING – II CREDIT: 05 (3-1-2)

Prerequisites: Geotechnical Engineering-I

UNIT 1. Stresses in Soils **06(L)**

Causes of stress in soil, Geostatic stress, Boussinesq's equation, Stress distribution diagrams, Vertical stress in soils under different types of loading, Newmark's influence chart, Westergard's equation.

UNIT 2. Earth Pressure and Retaining Walls **06(L)**

Types of lateral earth pressure, Rankine's and Coulomb's earth pressure theory, Graphical methods of determination of lateral earth pressures, Types of retaining walls, Design Principles, Stability conditions, Sheet pile walls.

UNIT 3. Bearing Capacity and Shallow Foundations **09(L)**

Methods of determining bearing capacity, Analytical methods of determining bearing capacity –Rankine's method, Terzaghi's theory, Meyerhof's method, Brinch–Hansen's method, Skempton's analysis, Vesic's analysis, IS code method, Settlement of foundation, Differential settlement, Settlement analysis, Bearing capacity based on tolerable settlement, Field tests, Types & design of shallow foundations.

UNIT 4. Pile & Well Foundation **09(L)**

Type of pile foundation, Types, Pile driving, Pile capacity by static and dynamic analysis, Pile load tests, Negative skin friction, Group action of piles, Settlement of pile groups, Under reamed piles – analysis and design.

Types of wells, Component of wells, Depth and bearing capacity of well foundation, Forces acting on well foundation, Construction and sinking of well foundation, Measures to prevent and rectify tilts and shifts.

UNIT 5. Machine Foundation **06(L)**

Types, Design criteria, Equation of motion, Natural frequency, Design of Reciprocating, Impact and Rotary type of machine foundation as per codal provisions.

List of Experiments:

1. Methods of Soil Exploration & Planning of Site investigations
2. Free swell Index & Differential free swell Index
3. Swelling pressure test
4. Static cone penetration test
5. Dynamic cone penetration test
6. Standard penetration test
7. Plate load test.
8. Vane shear test
9. Pile load test
10. Geophysical Investigations

References:

1. *J.E. Bowels: Foundation Analysis and Design, McGraw-Hill.*
2. *W. C. Teng: Foundation Design, Prentice-Hall.*
3. *B. M. Das: Principles of Foundation Engineering, PWS Publishing.*
4. *K.R. Arora: Soil Mechanics and Foundation Engineering, Standard Publishers Distributors.*
5. *P. C. Varghese: Foundation Engineering, PHI Learning Private Limited.*
6. *V. N. S. Murthy: Advance Foundation Engineering, CBS Publisher.*
7. *Nainan P. Kurian: Design of Foundation Systems, Narosa Publishing House.*
8. *SP 36-1: Compendium of Indian Standards on Soil Engineering: Part-1 Laboratory Testing of Soils for civil Engineering Purposes.*

CEXXXXX **WASTE WATER ENGINEERING** **CREDIT: 05 (3-1-2)**

Unit I: Wastewater collection & Conveyance **(6L)**

Wastewater collection: System of sanitation, estimation of wastewater flows and variations in wastewater flows. Estimation of storm water with numerical example. Flow in sewer: flow in fully and partially full sewer and design of sewer; Types of sewers, materials and construction of sewer, joints and sewer appurtenances, layout and construction of sewer lines, small bore sewer system.

Unit II: Introduction to Wastewater Treatment & its Kinetics **(6L)**

Objectives of wastewater treatment: characteristics, analysis of BOD, COD, solids and volatile solids & their significance, BOD kinetics, types of reactors and reactor analysis. Wastewater treatment, flow sheets for wastewater treatment. Kinetics of biological treatment system: biokinetics constants and their determination in both in batch and continuous systems

Unit III: Design of primary treatment units **(7L)**

Principles and design-Screens, grit chamber, equalization tank, primary settling tank

Unit IV: Aerobic treatment **(8L)**

Principles and design of aerobic treatment system: Suspended growth system- conventional activated sludge process and its modifications, oxidation ditch, Aerated lagoon, Waste stabilization pond. Principles and design of attached growth system-Trickling filter, bio-tower and rotating biological contactors. Microbiology of aerobic wastewater treatment systems.

Unit V: Anaerobic treatment & Sludge treatment **(9L)**

Principles and design of anaerobic wastewater treatment system including anaerobic filters, UASB reactor, Hybrid reactor etc. Sludge processing: saperation , sludge thickener, volume reduction, conditioning , aerobic and anaerobic sludge digestion. Nutrient removal, Nitrification and Denitrification process, Self purification of streams. Microbiology of aerobic wastewater treatment systems.

References:

1. Masters: Introduction to Environmental Engineering and Science, Prentice Hall Publication.
2. McGhee: Water supply and sewerage, Tata McgrawHill, publication.
3. Peavy, Rowe and Techbanoglous: Environmental Engineering, Tata McgrawHill, publication.
4. Garg S. K.: Environmental Enggineering (I&II), Khanna publication, New Delhi.
5. MetCalf & Eddy: Wastewater Engineering: Treatmment and Reuse, Tata McgrawHill, publication.
6. Mannual on Water Supply and Treatment, CPHEEO, Govt. of India.
7. Mannual on Sewerage and Sewage Disposal, CPHEEO, Govt. of India.

List of Experiments:

1. Determination of pH, electrical conductivity & turbidity of the given water sample.
2. Determination of acidity & alkalinity of given sample.
3. Determination of solids (TS, TSS, TDS, VSS and Inorganic solids) in a given wastewater sample.
4. Determination of optimum coagulant dose required for treating the given water sample.
5. Determination of chloride content & hardness in given water sample.
6. Determination of residual chlorine in the given water sample & the chlorine demand.
7. Determination of amount of iron present in the given sample.
8. Determination of amount of dissolved oxygen (DO) present in the sample.
9. Determination of amount of nitrate in the given water sample using PDA method.
10. Determination of amount of fluoride present in the given water sample.
11. Determination of chemical oxygen demand (COD) of given sample.
12. Determination of biochemical oxygen demand (BOD) of given sample.

References:

1. Standard Methods for the Examination of Water & Wastewater, APHA, AWWA, U.S.A.
2. Sawyer and McCarty: Chemistry for Environmental Engineering
3. Kotaiah: Environmental Engineering Laboratory Mannual, Charotar Publishing House. 36 Course Structure and Curriculum for B.Tech. P

CEXXXXX **CONCRETE STRUCTURES-I** **CREDIT: 04 (3-1-0)**

UNIT 1. Limit State Design **10 (L)**

Introduction to Reinforced Cement Concrete, Introduction to different design philosophy, Assumption, Distribution of Stresses on the cross section in bending, transformed area, Analysis and Design of a rectangular singly and doubly reinforced section, T and L sections for flexure, shear and bond.

UNIT 2. Behaviour of Section in Shear, Torsion , Bond **07(L)**

Behaviour of RC beam in shear, shear strength of beam with and without shear reinforcement, Minimum and Maximum shear reinforcement, Design of beam in shear using Limit state methods. Nature of bond between steel and concrete. Development of bond stress in reinforcement, Concept of development length and anchorage, Design of RC section in bond and calculation of development length using Limit state methods.

Failure of beam under torsion, interaction between shear and torsion and between moment and torsion, Concept of equivalent shear and moments. Analysis and design of beam curved in plan.

UNIT 3. Design of Slab & Staircase **05(L)**

Design of one-way and two-way solid slabs, and design of staircase.

UNIT 4. Design of Compression Members **07(L)**

Classification of Compression members, Effective length, Slenderness ratio and slenderness limit, Axially loaded short column's design - Limit State methods, Increase in permissible load in helically reinforced columns. Eccentrically loaded columns, Minimum eccentricity, Pu & Mu interaction diagrams, Design of Columns using IS-456 Design Aids with Uniaxial and Biaxial bending.

UNIT 5. Design of Footings **07(L)**

Structural behaviour of footings, design of footing for a wall and a single column, combined rectangular and trapezoidal footings, Design of strap footing.

References:

1. Fundamentals of Reinforced Concrete Structures , M L Gambhir , PHI

2. Reinforced Concrete Design , A K Jain , Nem Chand

3. Reinforced Concrete Design , P Dayratnam , Oxford IBH

4. Reinforced Concrete , Pillai & Menon , TMH 2

5. Reinforced Concrete Design , S N Sinha , TMH

CEXXXXX **IRRIGATION ENGINEERING** **CREDIT: 04 (3-1-0)**

UNIT 1 : General Considerations-Water Requirement of Crops **(10 L)**

Definition, Scope of subject, Advantage, Lift and flow irrigation, Development of irrigation in India, crops season, Important crops, Functions of water in plant growth, Soil moisture, consumptive use of water, Irrigation frequency, irrigation methods.

UNIT 2 : Canal Irrigation **(6 L)**

Classes of irrigation canal, Parts of a canal system, Preliminary survey, Detailed survey, Commanded areas, Channel alignment, Curves, Assessment of water requirement, Channel losses, Kennedy's Theory, Lacey's Theory, Longitudinal section, Schedule of area statistics and channel dimension cross- section of irrigation channel.

UNIT 3 : Water-Logging **(5 L)**

Definition, Effect, Causes and Anti-water logging measure, concrete lining, shotcrete lining, Asphaltic lining, Brick tile lining of earth material, Sections of lined channel. Drainage of Water logged land, Types of Drains, Open Drains, Closed Drains, Spacing of closed Drains.

UNIT 4 : Regulation and Control of Canal System **(5 L)**

Definition, requirement and classes of outlets, Non-modular outlet, Semi-module, Rigid module, Selection of outlet. Regulation, Measurement of discharge, Assessment of canal Revenue, Efficient Management of irrigation water.

UNIT 5 : Hydrologic Analysis and Design **(10L)**

Hydrologic statistics-probabilistic treatment of hydrologic data, frequency and probability functions, statistical parameters, probability distribution of hydrologic variables; Frequency analysis- extreme value distributions, flood frequency analysis, risk and reliability; Hydrologic design-design scale and level, safety factors and safety margins; uncertainty analysis.

Text/ Reference Books:

- *Singh, Dr. Bharat - Fundamentals of Irrigation Engineering, Nem Chand and Bros.*
- *Varshney, Dr. R.S., Gupta & Gupta - Theory and Design of Irrigation Structures Vol. I & II., Nem Chand and Bros.*
- *Punamia, Dr. B.C. and Pandey B.B. Lal, Irrigation and Water Power Engineering, Laxmi Publications(Pvt)Ltd.*
- *Bedient and Huber- Hydrology and Flodplain Analysis, Prentice Hall.*
- *Ojha,C.S.P. , Bhunya, P. and Berndtsson, R.- Engineering Hydrology, Oxford University Press Canada.*
- *Todd and Mays- Groundwater Hydrology, John Wiley and Sons, Inc.*
- *K. Subramanya - Engineering Hydrology, Tata McGraw Hill Education Pvt.Ltd.*
- *Modi, P.N. – Irrigation Water Resources and Water Power Engineering, Standard Book House*
- *Asawa, G.L. – Irrigation and Water Resources Engineering, New Age International.*
- *Walker, W.R. and Skogerboe, G.V. 1986. Surface irrigation theory and practice. Prentice-Hall, Inc.*

CEXXXXX **STRUCTURAL ANALYSIS- II** **CREDIT: 05 (3-1-2)**

UNIT 1. Force , Displacement and energy methods **08(L)**

Introduction, Force and Displacement Methods of Analysis of Indeterminate Structures, Method of Consistent Deformation for beams and plane frames. Method of Minimum Strain Energy for indeterminate beams, Trusses and plane frames.

UNIT 2. Moment Distribution and Slope Deflection Methods **07(L)**

Introduction, Moment distribution and Slope Deflection methods for continuous beams and plane frames.

UNIT 3. Matrix Method of Analysis **10(L)**

Introduction, Flexibility Method- Application to Beams, Trusses, Frames and Grid Structures; Stiffness Method- Application to Beams, Trusses, Frames and Grid Structures (including plane and space structures.

UNIT 4. Computer Oriented Direct Stiffness Method **07(L)**

Introduction, Application to Beams, Frames and Trusses.

UNIT 5. Plastic Analysis of Structures **04(L)**

Introduction, Analysis of Plastic Structures.

References:

1. Structural Analysis, Hibbeler, Pearson Publications.
2. Structural Analysis, Aslam Kassimali, Cengage Learning Publications.
3. Structural Analysis in Theory and Practice, Alan Williams, Elsevier Publications.
4. Elementary Structural Analysis, C. H. Norris, J. B. Wilbur and S. Utku., Tata Mcgraw Hill Publications.
5. Structural Analysis, L.S. Negi and R. S. Jangid, Tata Mcgraw Hill Publications.
6. Intermediate Structural Analysis, C. K. Wang, Tata Mcgraw Hill Publications.
7. Matrix Analysis of Framed Structures, W. Weaver (Jr.) and J. M. Gere, CBS Publications.

List of Experiments:

1. Analysis of Redundant Joint
2. Flexural Stiffness of Beam
3. Verification of Maxwell's Reciprocal Theorem
4. Analysis of Curved Members
5. Verification of Carry Over Factor
6. Three Hinged Arch
7. Two Hinged Arch
8. Elastically Coupled Beam
9. Unsymmetrical Bending

III Year (6th Semester)

CEXXXXX **STEEL STRUCTURES** **CREDIT: 05 (3-1-2)**

UNIT 1 : General Considerations **(6 L)**

General Considerations: Structural Steel, Stress-Strain Curve for Mild Steel, Rolled Steel Sections, Introduction to Plastic Analysis and Design, Working stress and Limit State Method of Steel Design

UNIT 2 : Connections **(6 L)**

Types of connections & joints, Bolted and Welded Connections, Load Transfer Mechanism, Failure Modes, Prying Action, Slip-Critical Connections, Moment Resistant Connections, Eccentric Connections, Beams-Column Connections, Bracket Connections, Framed Connections, Seat Connections.

UNIT 3 : Tension Members & Compression Members **(6 L)**

Types of Tension Members, Net and effective Sectional Areas, Types of Failure, Design Strength and design of Tension Member, Lug Angles, Splices, Gusset Plate. Effective Length and Slenderness Ratio of compression members, Classification of Cross Sections, Column Formula, Design of Axially Loaded Compression Members, Built-Up Columns (Latticed Columns), Encased Column, Column Splices, Design of Column Bases and Caps

UNIT 4 : Flexural Members & Plate Girders **(12 L)**

Structural behaviour of Beams, Types of Sections for flexural design Lateral Stability of Beams, Lateral Torsional Buckling, Plastic design of beams, Design of Laterally Supported Unsupported Beams Rolled Beams, Built-Up Beams, Lintels, Purlins, Bearing Plates, Design of Beam-columns

Elements of Plate Girder, General considerations, Proportioning of web, Proportioning of flanges, Design methods, End panel design, Design of Stiffeners, Curtailment of Flanges

UNIT 5: Gantry Girders & Industrial Buildings **(6 L)**

(a) Gantry Girders:-Introduction, Loads, Fatigue Effects, Design of gantry Girders.

(b) Industrial Buildings: - Introduction, Planning, Structural Framing, Types, Roof and Side Coverings, Elements of an Industrial Building, Design Steps of Industrial Building

References:

1. *Design of Steel Structures by N. Subramanian, Oxford University Press*
2. *Design of Steel Structures by K.S.Sai Ram,Pearson India Education Pvt. Ltd.*
3. *Design of Steel Structures by S. K. Duggal, Tata Mcgraw Hill.*
4. *Steel Structures by Robert Englekirk. Hohn Wiley & sons inc.*
5. *Structural Steel Design by Lambert Tall, Ronald Press Comp. New York*

BIS Codes of Practice

- *IS 800-2007 : Code of Practice for General Construction in Steel*
- *Hand Book for Structural Engineers : SP:6(1)-1964 [Reaffirmed in 2003]*
- *IS 808 : 1989 [Reaffirmed in 2003]*
- *Code of Practice for Design Loads (other than earthquake) for buildings and structures : IS 875 Part I to V : 1987 [Reaffirmed in 2003]*

List of Practical Drawing Assignments:

1. General Detailing Lay-out, details of Steel Sections and Symbolic representation of connections
2. Detailing of simple and moment resistant Bolted and Welded Connections
3. Detailing of trusses and Trussed Roofs: Using Angle sections, Tee sections, Channel sections, I-sections, and Tubular sections.
4. Detailing of beam and column connection using bolts and welds.
5. Detailing of beams , built-up columns and beam & column splicing
6. Detailing of column and footing connection using bolts and welds.
7. Detailing of Plate girder.
8. Detailing of Gantry girder.

Note: *Students are advised to submit at least two assignments, drawn using any standard drafting software such as AUTOCAD/Revit etc.*

Reference for Practical Drawing Assignments

- IS 800-2007: Code of Practice for General Construction in Steel
- SP 6(l)-1964: [Reaffirmed in 2003] Handbook for Structural Engineers 1. Structural Steel Sections
- IS 808-1989: [Reaffirmed 2004] Indian Standard Dimensions for Hot Rolled Steel Beam, Column, Channel and Angle Sections (Third Revision)
- IS 813-1986: [Reaffirmed 2003] Indian Standard Scheme of Symbols for Welding
- Kantimanthinathan, S. Manual of Detailing of Steel Structures IK International Publishing House Pvt. Ltd. New Delhi [2013]
- *AUTOCAD/Revit – Manuals*

CEXXXXX **CONCRETE STRUCTURES-II** **CREDIT: 04 (3-0-2)**

UNIT 1. Design of Flat & Circular Slabs **05(L)**

Design of flat slabs with and without drops. Circular slabs with various edge and loading conditions, and their usages.

UNIT 2. Design of Circular Water Tanks **05(L)**

Water tanks: Design criteria, material specifications and Permissible stresses for water retaining structures, Design of circular water tanks with fixed & flexible bases, situated on onground/underground.

UNIT 3. Design of Rectangular Water Tanks **05(L)**

Design of square /rectangular tanks situated on ground/underground, using approximate method and IS- code method.

UNIT 4. Design of Retaining Walls **06(L)**

Structural behaviour of retaining wall, stability of retaining wall against over-turning sliding and pressure developed under the base, Design of T-shaped retaining wall, Concept of counterfort retaining wall.

UNIT 5. Prestressed Concrete Design **15(L)**

Advantages of prestressing, methods of prestressing, Losses in prestress, analysis of simple prestressed rectangular and T-sections. Intoduction to design of elements, load balancing concept, profile of cable. Design of rectangular, T, I beam under flexure and shear using IS 1343

References:

1. Reinforced Concrete Design, Limit State Method ,Jain A.K. , NemChand
2. Reinforced Concrete Structures , M L Gambhir , PHI
3. Reinforced Concrete Design , Pillai & Menon , TMH
4. Design of Reinforced Concrete Structures , Dayaratnam , P. , Oxford IBH
5. Prestressed Concrete , Krishna Raju , N , CBS
6. IS 456-2000: Code of practice for plain & reinforced concrete.
7. SP-16: Design Aids of Reinforced Concrete to IS: 456-1978. 8. IS 3370-1968 & IS 3370-2009: Code of practice for water retaining structures
8. IS 1343 -2012 – Code of Practice for Prestressed Concrete

CONCRETE STRUCTURES-II (LAB.)

List of Practical in Structure Detailing Lab:

1. Simple Beam/Lintel
2. T-Beam floor
3. Rectangular Slabs
4. Brick wall and Isolated footing
5. Combined Rectangular and Trapezoidal Footing
6. Water tank
7. T. shape Retaining wall
8. Details of Flat slab & Circular Slabs
9. Details of Stair Case

Note - TWO hour practical is for drawing work (Manual & computer Aided).

CEXXXXX DESIGN OF HYDRAULIC STRUCTURES CREDIT: 04 (3-1-0)

UNIT 1: Principle of design irrigation hydraulic structures and Headworks- BLIGH'S Theory, Khosla's theory for determination of pressure and exit gradient, Hydraulic jump. Canal Head Works-Functions, Location, Layout of Head work, Weir, Canal head regulator, Design of Weirs on permeable foundation, silt control at head work. **10 (L)**

UNIT 2: Canal Regulation Works and Cross Drainage Works-Falls, Classification of falls, Design of falls, Distributory head regulator and cross-regulator, Escape, Bed bars. Necessity and types, Aquaduct, Syphon Aquaduct, Super passage, Canal syphon, Level crossing, Design of cross drainage work. **6 (L)**

UNIT 3: River Training Planning of Dams & Reservoirs- Objective, scope & classification of river & river training, stages, Methods of River Training, bank protection. Selection of Dam sites, Investigation, Estimation of storage capacity, Principle of Reservoir planning. Flood Routing, Reservior loss, Reservoir sedimentation. **6(L)**

UNIT 4: Types of Dams and Their Characteristics- Gravity Dams, Forces acting, method of analysis, Modes of failure and factors of safety, Elementary Profile of a gravity dam, Stability analysis, galleries. Earth dam, Foundation, Materials, Criteria for safe design, typical sections, compaction of Rock fill dam. Spilways, Spilway capacity, Types of spilway, Energy dissipation below spillway, Gates. **10(L)**

UNIT 5: Water Power-Hydro-Electric Power: Assessment of potential, Classification of power plants, Types of turbine, Power house. **4(L)**

Text/ Reference Books:

1. *Singh, Dr. Bharat - Fundamentals of Irrigation Engineering, Nem Chand and Bros.*
2. *Varshney, Dr. R.S., Gupta & Gupta - Theory and Design of Irrigation Structures Vol. I & II., Nem Chand and Bros.*
3. *Punamia, Dr. B.C. and Pandey B.B. Lal, Irrigation and Water Power Engineering, Laxmi Publications(Pvt)Ltd.*
4. *Modi, P.N. – Irrigation Water Resources and Water Power Engineering, Standard Book House*
5. *Bedient and Huber- Hydrlogy and Flodplain Analysis, Prentice Hall.*
6. *Asawa, G.L. – Irrigation and Water Resources Engineering, New Age International.*
7. *Walker, W.R. and Skogerboe, G.V. 1986. Surface irrigation theory and practice. Prentice-Hall, Inc.*

CEXXXXX SOFTWARE APPLICATION IN CIVIL ENGINEERING CREDIT: 02 (0-0-4)

(2 Lab Sessions of Two Hours in a week)

Objective:

With the advancement of the computer-based systems, software application are increasingly used to solve complex analysis and design problem in the different sub-specializations of Civil Engineering. This course is focused towards application of software for modeling, analysis and design in Civil Engineering problems. Moreover, students will become conversant pertaining to advantages and limitations of software, computation errors and validation of software results, failures due to errors in modeling, data entry and interpretation of software results.

List of Practical:

1. Develop the algorithms and computer programme to compute SFD and BMD of the beams (Cantilever, Simply Supported and Fixed) and slabs subjected to different kind of loading conditions.
2. Write computer programmes to design the different components of the RCC building structure using Limit State Method.
3. Analyse and design the multi-storey buildings as per relevant IS Codes using design software.
4. Prepare the calculations of concrete mixes design using spreadsheet.
5. Develop the algorithms and computer programme for designing the isolated, combined footing, raft footing and strip footing.
6. Carry out geometric design of roads using Road Design Software
7. Develop the computer programmes to design the flexible and rigid pavements using relevant IS Codes.
8. Develop the spatial database of a given area of interest using satellite data products and GIS Software.
9. Develop a computer program to design the sedimentation tank for water treatment.
10. Develop a computer program to calculate the runoff for a given catchment area.

References:

1. Krishnamoorthy, C. S., Rajeev, S., & Rajaraman, A. (2005). *Computer aided design: software and analytical tools*. Alpha Science International Limited.
2. Davenport, C., & Voiculescu, I. (2015). *Mastering AutoCAD Civil 3D 2016: Autodesk Official Press*. John Wiley & Sons.
3. Omura, George. *Mastering AutoCAD*. SYBEX Inc.
4. Raphael, B., & Smith, I. F. (2013). *Engineering informatics: Fundamentals of computer-aided engineering*. John Wiley & Sons.
5. Sharma, T.S., *Staad Pro V8i for Beginners: With Indian Examples,* Notion Press; 1st edition (1 January 2014)

CEXXXXX **ENGINEERING HYDROLOGY** **CREDIT: 03 (2-1-0)**

UNIT 1: General Consideration-Introduction, Definition, Hydrologic cycle. **4(L)**

UNIT 2: Precipitation and Abstractions-Precipitation, Infiltration and Evapotranspiration, Forms of precipitation, measurement, depth-area-duration and intensity-duration frequency relations, Evaporation - process, measurement, and estimation, Infiltration process, measurement, and estimation, Evapotranspiration measurement and estimation, Runoff, Rainfall Runoff correlations, Flow duration curve, Mass curve, Stream Flow measurements. **6(L)**

UNIT 3: Hydrograph- Factors affecting flow hydrograph, Unit hydrograph, its analysis, and S-curve hydrograph, Synthetic and instantaneous unit hydrographs. **4(L)**

UNIT 4: Floods-Statistical analysis, Flood frequency studies, Flood forecasting, Rational method, Time Area curves, Risk, reliability, and safety factor, Flood control measures. **6(L)**

UNIT 5: Flood Routing-Introduction to basic routing equations, Design flood, Channel and flood routing, Hydrologic Routing, Hydraulic routing. **4(L)**

References:

1. Engineering Hydrology by Dr. K. Subramanya Tata Mcgraw Hill Education Limited
2. Engineering Hydrology by Dr. C. S. P. Ojha

CEXXXXX **PROFESSIONAL ELECTIVE-I** **CREDIT: 03 (3-0-0)**

IV Year (7th Semester)

CEXXXXX RAILWAY & AIRPORT ENGINEERING CREDIT: 04 (3-1-0)

UNIT 1. History of Railways & Components of Permanent Way 06(L)

Introduction to Indian Railway; Classification of Railway Lines in India; Alignment of Railway Lines; Rail gauges; Coning of wheels; Permanent way- Rails, Rail Fastenings, Sleepers, Ballast.

.

UNIT 2. Geometric Design of Track 08(L)

Track stresses; Train Resistance & Tractive Power; Gradients; Grade compensation on curves, Track Geometry- Super elevation, Transition curves, Extra clearance on curves, Widening of gauges on curves; Points and Crossings- Switches, Components and types of crossing, Turnouts, Design of turnouts.

UNIT 3. Signalling and Stations and Yards 06(L)

Signalling: Classification, signalling systems, systems for controlling train movements, Interlocking. Classification of railway stations and yards; platforms, sidings.

UNIT 4. Modernization of Indian Railways 05(L)

High Speed Trains, Dedicated Freight Corridors, Rail Transit Systems, Safety & Information System

UNIT 5. Airport Engineering 11(L)

Air craft characteristics affecting airport design, Airport Planning, Runway Design. Airport Configuration- Runway configurations and operations, Taxiway configurations. Runway pavement design, design of overlay, Airport Layout, Runway lighting and Marking, Airport drainage.

References:

1. *Railway Engineering by Satish Chandra and M. M. Agarwal, Oxford University Press.*
2. *A Text Book of Railway Engineering by S. C. Saxena and S. P. Arora, Dhanpat Rai Publications.*
3. *Airport Planning and Design by S. K Khanna, M.G. Arora and S.S Jain,Nem Chand & Bros. Roorkee.*
4. *R. Horonjeff and F. X. Mckelvey, Planning & Design of Airports, 5th Edition, Mc Graw Hill.*

CEXXXXX EARTHQUAKE RESISTANT DESIGN CREDIT: 03 (2-1-0)

UNIT 1: Response of Single-Degree-Freedom Systems- Introduction to Seismology. Response of Structure to Earthquake motion, Modeling of structures, Dynamics of single degree of freedom system. Free and forced vibration, Response of the systems to general loadings. **4(L)**

UNIT 2: Response of Multi-Degree-Freedom Systems, Dynamics of multi degree of freedom system, Idealization of structures, Modal analysis of structures, Distributed mass system, Dynamics of soils and seismic response. Response spectra method. **5(L)**

UNIT 3: Concept of Structural Design-Conceptual design, Analysis of single and multi storey frame, Equivalent lateral force method. Design of Reinforced Concrete Buildings. Time history method, IS codes provisions. **5(L)**

UNIT 4: Design of Masonry and Steel Buildings-Design of Masonry buildings, Steel Buildings and non structural element, Material Properties, IS Code provisions for ductile design. **5(L)**

UNIT 5: Earthquake analysis and response of inelastic buildings. Approximate analysis procedures, Uncoupled modal response history analysis, Modal pushover analysis, Earthquake dynamic of base isolated buildings. **5(L)**

References:

1. Introduction to Structural Dynamics - J.M. Biggs, McGraw-Hill Companies.
2. Earthquake Resistant of Design of structures, S.K.Duggal, Oxford University Press.
3. IS: 1893 – 2002, Criterion for Earthquake Resistant Design.
4. Engineering Vibrations - L.S. Jacobsen & R.S. Ayre, McGrawHill Book Co., New York
5. Structural Dynamics - Theory & computation - Mario Paz, Springer, 1997.
6. Dynamics of Structures Theory and Applicaions to Earthquake Engineering - Anil K. Chopra, Pearson/Prentice Hall.
7. Structural dynamics - R. Roy Craig Jr, Wiley, 1981.
8. Dynamics of structures - R. W. Clough and J Penjien.McGraw-Hil, Inc. 19

CEXXXXX PROFESSIONAL ELECTIVE-II CREDIT: 03 (3-0-0)